

Covariance + Correlation:

In settings with at least 2 different r.v.'s, it is often of interest to quantify how a pair of r.v.'s might vary with each other.

For instance: X represents the # of coal plants in an area and Y is the prevalence of lung cancer.

→ Is a location with high value of X likely to have a high or low value of Y ?

Def. (Covariance): For joint r.v.'s X, Y , their covariance, $\text{Cov}(X, Y)$, equals:

$$\text{Cov}(X, Y) = E\left[\underbrace{(X - E[X])(Y - E[Y])}\right]$$

↑ this is positive for (x, y) satisfying either:
i) $x > EX$ and $y > EY$
ii) $x < EX$ and $y < EY$

Note: Covariance is a linear measure of how strongly X, Y are related.

Properties:

- $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$
- $\text{Cov}(X, X) = \text{Var}(X)$
- If X and Y are independent, then $\text{Cov}(X, Y) = 0$
- For any constants $a, b, c, d \in \mathbb{R}$,
$$\text{Cov}(aX + b, cY + d) = a \cdot c \cdot \text{Cov}(X, Y)$$
- For any joint r.v.'s $Y, X_1, X_2, \dots, X_n$, we have:
$$\text{Cov}(Y, X_1 + X_2 + \dots + X_n) = \text{Cov}(Y, X_1) + \dots + \text{Cov}(Y, X_n)$$

Proof: Last property, for $n=2$

$$\begin{aligned}
 \text{Cov}(Y, X_1 + X_2) &= E[(Y - EY)(X_1 + X_2 - E[X_1 + X_2])] \\
 &= E[(Y - EY)(X_1 - EX_1) + (Y - EY)(X_2 - EX_2)] \quad \text{distribute the product} \\
 &= E[(Y - EY)(X_1 - EX_1)] + E[(Y - EY)(X_2 - EX_2)] \quad \text{linearity of expectation} \\
 &= E[(Y - EY)(X_1 - EX_1)] + E[(Y - EY)(X_2 - EX_2)] \quad \text{by def.} \\
 &= \text{Cov}(Y, X_1) + \text{Cov}(Y, X_2) \quad \square
 \end{aligned}$$

Example:

$p(x, y)$		x		Marginals:			
y	$p(x, y)$	-1	+1		x	-1	+1
	-1	0	$\frac{1}{4}$	$p_x(x)$	$\frac{1}{2}$	$\frac{1}{2}$	
	0	$\frac{1}{2}$	0	y	-1	0	+1
	+1	0	$\frac{1}{4}$	$p_y(y)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

Compute:

- $E[X] = (-1) \cdot \frac{1}{2} + (+1) \left(\frac{1}{4} + \frac{1}{4} \right) = 0$
- $E[Y] = (-1) \cdot \frac{1}{4} + 0 \left(\frac{1}{2} \right) + (+1) \left(\frac{1}{4} \right) = 0$
- $$\begin{aligned}
 \text{Cov}(X, Y) &= E[X \cdot Y] - E[X]E[Y] \\
 &= \left[(1 \cdot -1) \cdot \frac{1}{4} + (-1 \cdot 0) \cdot \frac{1}{2} + (1 \cdot 1) \cdot \frac{1}{4} \right] - 0 \cdot 0 \\
 &= \left[-\frac{1}{4} + \frac{1}{4} \right] - 0 = 0
 \end{aligned}$$

But clearly X, Y are not independent!

e.g. $p(-1, +1) = 0 \neq p_x(-1) p_y(+1) = \frac{1}{8}$

So $\text{Cov}(X, Y) = 0$ does not imply that X, Y are independent...

If 2 random variables are not independent, then we can still express the variance of their sum.

Prop.: For random variables X, Y , we have:

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2 \cdot \text{Cov}(X, Y)$$

Proof: From $\textcircled{4}$ above,

$$\begin{aligned} \text{Var}(X+Y) &= \mathbb{E}[(X - \mathbb{E}[X])^2] + \mathbb{E}[(Y - \mathbb{E}[Y])^2] \\ &\quad + 2 \cdot \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \\ &= \text{Var}(X) + \text{Var}(Y) + 2 \cdot \text{Cov}(X, Y) \quad \text{by def'n.} \quad \square \end{aligned}$$

Example: If $Y = -X$, then:

$$\bullet \text{Var}(Y) = \text{Var}(-1 \cdot X) = (-1)^2 \cdot \text{Var}(X) = \text{Var}(X)$$

$$\bullet \text{Cov}(X, Y) = \text{Cov}(X, -1 \cdot X) = (-1) \cdot \text{Cov}(X, X) = -\text{Var}(X)$$

$$\Rightarrow \text{Var}(X+Y) = \text{Var}(X) + \text{Var}(X) + 2 \cdot (-\text{Var}(X))$$

$$= \underline{0}$$

$\leadsto$ Why? $X+Y = X+(-X) = 0$, which is not random.

Hence, it has zero variance.

A problem with covariance - it depends on the units of X, Y .

Ex: $X = \text{height, in.}$ $\leftrightarrow$ $Z = \text{height, cm}$
 $Y = \text{midterm score.}$

By Y^{th} property above, $\text{Cov}(Z, Y) = 2.54 \text{Cov}(X, Y)$
since $Z = 2.54 \frac{\text{cm}}{\text{in}} \cdot X$.

$\leadsto$ But Z, Y are not any more strongly related than X, Y ...

Ideally, the units shouldn't affect our measure for the strength of a relationship. So we turn to a second metric.

Def. (Correlation): The correlation between X, Y , denoted $\rho(X, Y)$, is:

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}$$

Properties:

- $-1 \leq \rho(X, Y) \leq +1$
- For any constants $a, b > 0$, $\rho(a \cdot X, b \cdot Y) = \rho(X, Y)$
- If X, Y are independent, then $\rho(X, Y) = 0$.

Note: The magnitude of $\rho(X, Y)$ tells us the strength of their relationship
 $\rho(X, Y) > 0$ means X, Y are positively correlated.

$\rho(X, Y) < 0$ means X, Y are negatively correlated.

If $|\rho(X, Y)| = 1$, then X, Y are perfectly correlated, and it must be that $Y = a \cdot X + b$ for some constants a, b .

Takeaway: Correlation is unitless! And can now be compared for different pairs of r.v.'s. If $|\rho(X, Y)| > |\rho(Z, Y)|$, then we would say that X and Y have a "stronger" relationship than Z and Y .

Conditional Expectations:

Recall that conditional pmf/pdf's study how the distribution of one r.v. depends on what the other r.v. equals.

With jointly distributed r.v.'s, we can also ask about the expected value of one, e.g. X , conditioned on another, e.g. $Y=y$.

Recall the conditional pdf/pmf is defined as

$$p_{X|Y}(x|y) = \frac{p(x,y)}{p_Y(y)}, \quad f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)}$$

We use these to compute conditional expectations

Def. (Conditional Expectation): If X, Y are joint discrete r.v.'s, then

$$E[X|Y=y] = \sum_{x \in \text{Range}(X)} x \cdot p_{X|Y}(x|y)$$

If X, Y are continuous, then:

$$E[X|Y=y] = \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx$$

Note: The only difference here from just $E[X]$ is that we use the conditional pmf/pdf instead of the (marginal) pmf/pdf of X .

Notice that these conditional expectations depend on which value y we plug in. I.e. $E[X|Y=y]$ is something that depends on y itself. But the true value of Y is random - it may or may not equal a particular y .

Example:

$p(x,y)$		x	
		0	1
y	0	$1/8$	$3/8$
	1	$3/8$	$1/8$

Compute the conditional pmf of X given Y :

$$p_Y(0) = \frac{1}{2}, \quad p_Y(1) = \frac{1}{2}$$

		x	
		0	1
	$p_{X Y}(x 0)$	$1/4$	$3/4$
	$p_{X Y}(x 1)$	$3/4$	$1/4$

So:

$$E[X|Y=0] = 0 \cdot \frac{1}{4} + 1 \cdot \frac{3}{4} = \underline{\underline{\frac{3}{4}}}$$

$$E[X|Y=1] = 0 \cdot \frac{3}{4} + 1 \cdot \frac{1}{4} = \underline{\underline{\frac{1}{4}}}$$

Notice that $p_X(0) = p_X(1) = 1/2$, so $E[X] = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 1 = \underline{\underline{\frac{1}{2}}}$

But also note that:

$$\begin{aligned} & p_Y(0) \cdot E[X|Y=0] + p_Y(1) \cdot E[X|Y=1] \\ &= \frac{1}{2} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{1}{4} = \underline{\underline{\frac{1}{2}}} \end{aligned}$$

$\leadsto$ Is it a coincidence that this equals $E[X]$?
No!

Prop. (Tower Rule of Conditional Expectation): For any joint r.v.'s

X, Y , we have:

$$E[\underline{E[X|Y]}] = E[X]$$

$\uparrow$ this is now a random quantity, depending on what value Y takes.

e.g. if Y is discrete, then:

$E[X|Y]$ equals $E[X|Y=y]$ with prob. $p_Y(y)$

Proof: Discrete Case

$$E[E[X|Y]] = \sum_{y \in \text{Range}(Y)} E[X|Y=y] \cdot p_Y(y)$$

since $E[X|Y]$ depends only on the value of Y

by def. of $E[X|Y=y]$

$$= \sum_{y \in \text{Range}(Y)} \left(\sum_{x \in \text{Range}(X)} x \cdot p_{X|Y}(x|y) \right) p_Y(y)$$

bring $p_Y(y)$ into 2nd summation

$$= \sum_{y, x} x \cdot p_{X|Y}(x|y) p_Y(y)$$

by def. of $p_{X|Y}(x|y)$

$$= \sum_{y, x} x \cdot \frac{p(x, y)}{p_Y(y)} \cdot p_Y(y)$$

$$= \sum_{y, x} x \cdot p(x, y) = E[X] \quad \square$$

$$= \sum_x x \cdot \underbrace{\left[\sum_y p(x, y) \right]}_{= p_X(x)}$$

$$= \sum_x x \cdot p_X(x) = E[X]$$